

# Chenlai Shi

<https://shichenlai.github.io/>

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## SUMMARY

AI Software Engineer with strong full-stack and applied machine learning experience building intelligent, production-grade applications. Experienced in integrating LLMs and deep learning models into scalable backend services and interactive web platforms. Proven expertise in delivering end-to-end features—from model development and evaluation to deployment, monitoring, and continuous improvement—across cloud and distributed systems.

## TECHNICAL SKILLS

|                |                                                           |
|----------------|-----------------------------------------------------------|
| Languages      | Python, JavaScript, SQL, R, Linux, Git                    |
| AI/ML & GenAI  | PyTorch, TensorFlow, OpenAI APIs, LangChain/LangGraph     |
| MLOps & DevOps | Kubernetes, OpenShift, Docker, CI/CD, ONNX                |
| Backend & Data | FastAPI, Flask, Node.js, Redis, Nginx, Solr, Cloud, Spark |

## EXPERIENCE

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|------------------------------------------------------------------------------|-------------------|
| UT Southwestern Medical Center                                               | 06/2020 - Present |
| Senior Platform Engineer – <i>AI workflow systems and HPC infrastructure</i> | 02/2025 - Present |

- Designed and developed **Astrocyte**, a production AI-enabled workflow platform supporting 500+ users and 10,000+ runs, delivering scalable backend services and interactive web-based tools for research applications.
- Integrated a **context-aware multimodal RAG assistant** (LLM-based) using FastAPI, LangChain/LangGraph, FAISS, Ollama, and MySQL, supporting session-aware chat, async task processing, hallucination/answer grading, web fallback, scraped domain knowledge ingestion, and image, PDF, audio ASR/TTS workflows.
- Developed robust application capabilities including workflow composition, execution tracking, data search, and ranking features across distributed compute environments. Built analytics and monitoring tools for compute/storage utilization (122M+ CPU hours, 108M+ GPU hours), supporting performance tuning, retraining workflows, and system observability.
- Optimized model performance and system reliability in production, improving workflow success rate from 43.7% to 80.2% through monitoring, debugging, and orchestration improvements.

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|-----------------------------------------------------------------|-------------------|
| Data Scientist – <i>Machine learning systems and applied AI</i> | 06/2020 - 02/2025 |
|-----------------------------------------------------------------|-------------------|

- Architected, trained, and optimized high-performance transformer models in **PyTorch** for long-sequence tasks; implemented custom embedding layers and memory-efficient attention mechanisms to scale training across multi-GPU clusters.
- Engineered and deployed **Bay Server**, a production-grade model-serving platform using **Flask, Redis, and RQ (Redis Queue)**; unified complex deep learning models under standardized REST APIs to boost IO throughput.
- Led the end-to-end full-stack development of **TSS Analyzer**, a high-throughput data application integrating **ONNX runtime** for accelerated model inference, **Solr** for scalable search indexing.
- Spearheaded full-stack engineering of an interactive genomic sequence browser using **React, Node.js, and Flask**, implementing robust server-side caching and data pre-fetching to smoothly render intensive multi-track visualizations.
- Optimized **PyTorch** inference pipelines by profiling execution bottlenecks, reducing memory footprint, and exporting models to ONNX Runtime for low-latency, high-throughput production workloads.

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|---------------------------------------|-------------------|
| Dean Health Plan, Inc.                | 02/2019 - 05/2020 |
| Analyst – <i>Healthcare analytics</i> |                   |

- Developed ensemble models on 1M+ medical claims and EHR records using **Scikit-learn**, improving patient outcomes and driving financial savings: \$5400 per patient, 0.2 fewer admissions, and 0.16 fewer ER visits.

## PUBLICATION

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|------------------------------------------------------------------------------------|--------------|
| <a href="#">Dirichlet Diffusion Score Model for Biological Sequence Generation</a> | ICML 2023    |
| <a href="#">Sequence Basis of Transcription Initiation in Human Genome</a>         | Science 2023 |

## EDUCATION

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|--------------------------------------------------------------------|-------------------|
| Master of Science in Statistics                                    | 09/2017 - 12/2018 |
| Master of Science in Economics                                     | 09/2014 - 05/2017 |
| University of Wisconsin-Madison                                    | GPA: 3.6          |
| Bachelor of Science in Economics and Finance, Minor in Mathematics | 09/2010 - 05/2014 |
| Hong Kong University of Science and Technology                     | GPA: 3.9          |
| First Class Honors, HKSAR Government Scholarship                   |                   |